

ASSIGNMENT 4 SOLUTIONS

QUESTION – 1 - Print Numbers from 1 to N

```
import java.util.Scanner;

class printnNumbers
{
    static void nNumbers(Scanner sc) {

        System.out.print("Enter a number: ");

        int num = sc.nextInt();

        for(int i=1; i<=num; i++) {
            System.out.println(i);
        }
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        nNumbers(sc);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac printnNumbers.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java printnNumbers
Enter a number: 7
1
2
3
4
5
6
7
```

QUESTION – 2 - Print Multiples of 3 between 1 and N

```
import java.util.Scanner;

class multipleOf3
{
    void multiple(Scanner sc) {
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        for(int i=1; i<=num; i++) {
            if(i % 3 == 0){
                System.out.println(i);
            }
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        multipleOf3 obj = new multipleOf3();
        obj.multiple(sc);
        sc.close();
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac multipleOf3.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java multipleOf3
Enter a number: 40
3
6
9
12
15
18
21
24
27
30
33
36
39
```

QUESTION – 3 - Calculate the Factorial of a Number

```
import java.util.Scanner;

class Factorial {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int n = sc.nextInt();

        int factorial = 1;

        for (int i = 1; i <= n; i++) {

            factorial = factorial * i;

        }

        System.out.println("Factorial of " + n + " is " + factorial);

        sc.close();

    }

}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac Factorial.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java Factorial
Enter a number: 5
Factorial of 5 is 120
```

QUESTION – 4 - Print Even Numbers from 1 to N

```
import java.util.Scanner;

class evenNumbers
{
    static void printEvenNumbers(Scanner sc) {

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        for(int i=1; i<=num; i++) {
            if (i % 2 == 0) {
                System.out.println(i);
            }
        }
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        printEvenNumbers(sc);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac evenNumbers.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java evenNumbers
Enter a number: 20
2
4
6
8
10
12
14
16
18
20
```

QUESTION – 5 - Sum of Odd Numbers between 1 and N

```
import java.util.Scanner;

class sumOfOddNumbers{

    static int printSumOfOddNumbers(Scanner sc) {

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int sum = 0;
        for(int i=1; i<=num; i++) {
            if(i % 2 != 0) {
                sum = sum + i;
            }
        }
        return sum;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        int result = printSumOfOddNumbers(sc);

        System.out.println("Sum of Odd numbers = " + result);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac sumOfOddNumbers.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java sumOfOddNumbers
Enter a number: 40
Sum of Odd numbers = 400
```

QUESTION – 6 - Print All Elements of an Array

```
import java.util.Scanner;

class forEachLoop
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 numbers: ");
        for(int i=0; i<5; i++) {
            numbers[i] = sc.nextInt();
        }

        System.out.println("The numbers you have entered are: ");

        for(int number: numbers){
            System.out.println(number);
        }
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac forEachLoop.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java forEachLoop
Enter 5 numbers:
1
2
3
4
5
The numbers you have entered are:
1
2
3
4
5
```

QUESTION – 7 - Find the Sum of All Elements in an Array

```
import java.util.Scanner;

class sumOfIntegerinArray
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 numbers: ");

        for(int i=0; i<5; i++) {
            numbers[i] = sc.nextInt();
        }

        int sum = 0;

        for(int number: numbers){
            sum = number + sum;
        }

        System.out.println("Sum = " + sum);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac sumOfIntegerinArray.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java sumOfIntegerinArray
Enter 5 numbers:
1
2
3
4
5
Sum = 15
```

QUESTION – 8 - Print All Names in a String Array

```
import java.util.Scanner;

class namesInStringArray
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String[] names = new String[5];

        System.out.println("Enter 5 names: ");

        for(int i=0; i<5; i++) {
            names[i] = sc.nextLine();
        }

        System.out.println("The names are : ");

        for(String name: names){
            System.out.println(name);
        }

    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac namesInStringArray.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java namesInStringArray
Enter 5 names:
Aditee
Anjali
Ashok
Kishor
Pratibha
The names are :
Aditee
Anjali
Ashok
Kishor
Pratibha
```


QUESTION 9 - Find the Largest Element in an Array

```
import java.util.Scanner;

import java.util.Arrays;

class largestNumber
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int numbers[] = new int[5];

        System.out.println("Enter 5 numbers: ");

        for(int i=0; i<5; i++) {
            numbers[i] = sc.nextInt();
        }

        Arrays.sort(numbers);

        int largest = numbers[4];

        System.out.println("The largest element is: " + largest);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac largestNumber.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java largestNumber
Enter 5 numbers:
9
3
8
2
1
The largest element is: 9
```

QUESTION – 10 - Find the Average of Elements in an Array

```
import java.util.Scanner;

class average
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int numbers[] = new int[5];
        System.out.println("Enter 5 numbers: ");

        for(int i=0; i<5; i++) {
            numbers[i] = sc.nextInt();
        }

        int sum=0;

        for(int number: numbers) {
            sum = number + sum;
        }

        double result = sum/5;
        System.out.println("The average is: " + result);
        sc.close();
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac average.java
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java average
```

```
Enter 5 numbers:
```

```
10
```

```
20
```

```
30
```

```
40
```

```
50
```

```
The average is: 30.0
```

QUESTION – 11 - Count Positive and Negative Numbers in an Array

```
import java.util.Scanner;

class positiveAndNegative
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[6];

        int positive = 0;
        int negative = 0;

        System.out.println("Enter 6 numbers: ");

        for(int i=0; i<6; i++) {
            numbers[i] = sc.nextInt();

            if(numbers[i] > 0) {
                positive++;
            }

            else if(numbers[i] < 0) {
                negative++;
            }
        }

        System.out.println("Positive numbers: " + positive);
        System.out.println("Negative numbers: " + negative);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac positiveAndNegative.java

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java positiveAndNegative
Enter 6 numbers:
12
-89
-76
-34
12
-57
Positive numbers: 2
Negative numbers: 4
```

QUESTION – 12 - Sort an Array in Ascending Order

```
import java.util.Scanner;

import java.util.Arrays;

class ascendingOrder
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter any 5 numbers: ");

        for(int i=0; i<5; i++) {

            numbers[i] = sc.nextInt();
        }

        Arrays.sort(numbers);

        System.out.println("Sorted array: ");
```

```

    for(int i=0; i<5; i++) {

        System.out.println(numbers[i] + " ");

    }

    sc.close();

}
}

```

```

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java ascendingOrder
Enter any 5 numbers:
9
6
3
4
1
Sorted array:
1
3
4
6
9

```

QUESTION – 13 - Check if an Array Contains a Specific Element

```

import java.util.Scanner;

import java.util.Arrays;

class SearchArray
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Integer[] numbers = new Integer[5];
    }
}

```

```
System.out.println("Enter any 5 numbers: ");
```

```
for(int i=0; i<5; i++) {  
    numbers[i] = sc.nextInt();  
}
```

```
System.out.println("Enter a number to search: ");  
int search = sc.nextInt();
```

```
if(Arrays.asList(numbers).contains(search)) {  
    System.out.println("Found");  
}  
else {  
    System.out.println("Not found");  
}
```

```
}
```

```
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac SearchArray.java
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java SearchArray
```

```
Enter any 5 numbers:
```

```
1
```

```
2
```

```
3
```

```
4
```

```
5
```

```
Enter a number to search:
```

```
9
```

```
Not found
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac SearchArray.java
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java SearchArray
```

```
Enter any 5 numbers:
```

```
1
```

```
2
```

```
3
```

```
4
```

```
5
```

```
Enter a number to search:
```

```
3
```

```
Found
```

QUESTION – 14 - Find the Index of an Element in an Array

```
import java.util.Scanner;

import java.util.Arrays;

class indexOfAnElement
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter any 5 numbers: ");

        for(int i=0; i<5; i++) {
            numbers[i] = sc.nextInt();
        }

        System.out.println("Enter a number to search: ");
        int search = sc.nextInt();

        int result = Arrays.binarySearch(numbers, search);

        if(result >= 0) {
            System.out.println("The number " + search + " is found at index " + result);
        } else {
            System.out.println("Not found");
        }
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac indexOfAnElement.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java indexOfAnElement
Enter any 5 numbers:
1
2
3
4
5
Enter a number to search:
1
The number 1 is found at index 0
```

QUESTION – 15 - Write a program to print the following pattern

```
import java.util.Scanner;

class pattern
{
    public static void main(String[] args) {

        for(int i=1; i<=5; i++) {

            for(int j=1; j<=i; j++) {

                System.out.print(i);

                if(j<i) {
                    System.out.print("*");
                }
            }

            System.out.println();
        }
    }
}
```



```

for(int i=5; i>=2; i--) {

    for(int j=1; j<=i; j++) {

        System.out.print(i);

        if(j<i) {
            System.out.print("*");
        }

    }

    System.out.println();
}
}

```

```

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern.java

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2

```

QUESTION – 16 - Write a program to print the following pattern

```

import java.util.Scanner;

class pattern1
{
    public static void main(String[] args) {

```

```

for(int i=1; i<=5; i++) {

    for(int j=1; j<=i; j++) {

        System.out.print(i);

        if(j<i) {

            System.out.print("*");

        }

    }

    System.out.println();

}

}

```

```

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern1.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern1
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5

```

QUESTION – 17 - Write a program to print the following pattern

```

import java.util.Scanner;

class pattern2
{
    public static void main(String[] args) {

        for(int i=1; i<=5; i++) {

```

```

        for(int j=1; j<=i; j++) {

            System.out.print(2 * j - 1);

            if(j<i) {

                System.out.print("*");

            }

        }

        System.out.println();

    }

}

```

```

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern2.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern2
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9

```

QUESTION – 18 - Write a program to print the following pattern

```

import java.util.Scanner;

class pattern3
{
    public static void main(String[] args) {

        for(int i=1; i<=5; i++) {

            for(int j=1; j<=5; j++) {

                System.out.print(i);

```

```

        }

        System.out.println();

    }

}

}

```

```

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern3.java

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern3
11111
22222
33333
44444
55555

```

QUESTION – 19 - Write a program to print the following pattern

```

import java.util.Scanner;

class pattern4
{
    public static void main(String[] args) {

        for(int i=1; i<=5; i++) {

            for(int j=1; j<=i; j++) {

                System.out.print(i);

            }

            System.out.println();

        }

    }

}

```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern4.java  
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern4  
1  
22  
333  
4444  
55555
```

QUESTION – 20 - Write a program to print the following pattern

```
import java.util.Scanner;  
  
class pattern5  
{  
    public static void main(String[] args) {  
  
        for(int i=1; i<=5; i++) {  
  
            for(int j=1; j<=i; j++) {  
  
                System.out.print(j);  
            }  
            System.out.println();  
        }  
    }  
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern5.java  
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern5  
1  
12  
123  
1234  
12345
```

QUESTION – 21 - Write a program to print the following pattern

```
import java.util.Scanner;

class pattern6
{
    public static void main(String[] args) {

        int number = 1;

        for(int i=1; i<=5; i++) {

            for(int j=1; j<=i; j++) {

                System.out.print(" " + number);

                number++;

            }

            System.out.println();

        }

    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern6.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern6
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

QUESTION – 22 - Write a program to print the following pattern

```
import java.util.Scanner;

class pattern7
{
    public static void main(String[] args) {

        for(int i=1; i<=6; i++) {

            for(int j=1; j<=6; j++) {

                if(i==1 || i==6 || j==1 || j==6) {

                    System.out.print("*");
                }
                else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern7.java
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern7
```

```
*****
*      *
*      *
*      *
*      *
*****
```

QUESTION – 23 - Write a program to print the following pattern

```
import java.util.Scanner;

class pattern8
{
    public static void main(String[] args) {

        for(int i=1; i<=5; i++) {

            for(int j=1; j<=5-i; j++) {
                System.out.print(" ");
            }

            for(int j=1; j<=2*i-1; j++) {
                System.out.print("*");
            }
            System.out.println();
        }

        for(int i=4; i>=1; i--) {

            for(int j=1; j<=5-i; j++) {
                System.out.print(" ");
            }

            for(int j=1; j<=2*i-1; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```



```
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac pattern8.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java pattern8
  *
 ***
*****
*****
*****
*****
*****
 ***
  *
```

QUESTION – 24 - Reverse a String

```
import java.util.Scanner;

class reverseString
{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a String: ");
        String str = sc.nextLine();

        String reversed = new StringBuilder(str).reverse().toString();
        System.out.println(reversed);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java reverseString.java
Enter a String: hello
olleh

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac reverseString.java

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java reverseString
Enter a String: Aditee
eetida
```

QUESTION – 25 - Count Vowels in a String

```
import java.util.Scanner;

class countOfVowels
{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a String: ");
        String str = sc.nextLine();

        int count = 0;

        for(int i=0; i<str.length(); i++) {

            char ch = str.charAt(i);

            if(ch == 'a' || ch=='e' || ch=='i' || ch=='o' || ch=='u'){
                count++;
            }
        }

        System.out.println("Number of Vowels in " + str + " are: " + count);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac countOfVowels.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java countOfVowels
Enter a String: programming
Number of Vowels in programming are: 3
```

QUESTION – 26 - Check if a String is a Palindrome

```
import java.util.Scanner;

class Palindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String clean = str.replaceAll("[^a-zA-Z0-9]", "");

        String reverse = new StringBuilder(clean).reverse().toString();

        if (clean.equalsIgnoreCase(reverse)) {
            System.out.println("The string '" + str + "' is a palindrome.");
        }
        else {
            System.out.println("The string '" + str + "' is not a palindrome.");
        }

        sc.close();
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac Palindrome.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java Palindrome
Enter a string: madam
The string 'madam' is a palindrome.
```

QUESTION – 27 - String Literal and Object Creation

```
class StringObject {  
    public static void main(String[] args) {  
  
        String str1 = "hello";  
        String str2 = "hello";  
  
        System.out.println("Both variables point to the same object: " + (str1 == str2));  
    }  
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac StringObject.java  
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java StringObject  
Both variables point to the same object: true
```

QUESTION – 28 - String Creation with new Keyword

```
class StringComparison {  
    public static void main(String[] args) {  
        String str1 = new String("hello");  
        String str2 = new String("hello");  
        System.out.println("Using == : " + (str1 == str2));  
        System.out.println("Using .equals(): " + str1.equals(str2));  
    }  
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac StringComparison.java  
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java StringComparison  
Using == : false  
Using .equals(): true
```

QUESTION – 29 - String Concatenation and Object Creation

```
class StringConcatenation {  
    public static void main(String[] args) {  
  
        String str1 = "hello";  
        String str2 = "world";  
  
        String str3 = str1 + str2;  
  
        System.out.println("Is str3 pointing to the same object as str1? " + (str3 == str1));  
    }  
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac StringConcatenation.java  
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java StringConcatenation  
Is str3 pointing to the same object as str1? false
```

QUESTION – 30 - String Pool with intern() Method

```
public class StringIntern {  
    public static void main(String[] args) {  
  
        String str1 = new String("hello");  
  
        String str2 = str1.intern();  
  
        String str3 = "hello";  
  
        System.out.println("Is str2 and str3 pointing to the same object? " + (str2 == str3));  
    }  
}
```

```
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac StringIntern.java  
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java StringIntern  
Is str2 and str3 pointing to the same object? true
```

QUESTION – 31 - Multiple String Literals with Same Content

```
class StringLiteral {  
    public static void main(String[] args) {  
  
        String str1 = "java";  
        String str2 = "java";  
        String str3 = "java";  
  
        System.out.println("All strings point to the same object: " +  
            (str1 == str2 && str2 == str3));  
    }  
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>javac StringLiteral.java  
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java StringLiteral  
All strings point to the same object: true
```